QUESTION 1. **(10 Marks)**

For a matrix $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(\boldsymbol{A}) = a_{i1} + a_{i2} + \ldots + a_{in}\,.$$

In other words, $r_i(\boldsymbol{A})$ is the sum of entries in the ith row of $\boldsymbol{A}$.

(a) Consider the set $\mathcal{U} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) = 1 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{U}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

(b) Consider the set $\mathcal{V} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) \geq 0 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

(c) Consider the set $\mathcal{W} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) = 0 \text{ for some } i = 1, 2, \ldots, n\}$. Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

QUESTION 2. **(20 Marks)**

Let $\boldsymbol{T} : \mathcal{P}_4(\mathbb{R}) \to \mathbb{R}^3$ be the map defined by

$$\boldsymbol{T}(f) = (f(0), f(1), f(2))\,.$$

(a) Show that $\boldsymbol{T}$ is a linear map.

(b) Consider the following basis $\mathcal{B} = \{1, x, x^2, x^3, x^2(x-1)(x-2)\}$ for $\mathcal{P}_4(\mathbb{R})$ (you need not show that $\mathcal{B}$ is a basis). Let $\mathcal{A} = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$ be the standard basis for $\mathbb{R}^3$. Write down the matrix representation of $\boldsymbol{T}$ with respect to $\mathcal{B}$ and $\mathcal{A}$.

(c) Determine the dimensions of null($\boldsymbol{T}$) and range($\boldsymbol{T}$). In your solution, you *must* provide a basis for either null($\boldsymbol{T}$) and range($\boldsymbol{T}$). You need *not* provide bases for both null($\boldsymbol{T}$) and range($\boldsymbol{T}$).

(d) Determine whether the following properties of $\boldsymbol{T}$ is true. You are to explain your answer.

  (i) Is $\boldsymbol{T}$ is injective?

  (ii) Is $\boldsymbol{T}$ is surjective?

  (iii) Is $\boldsymbol{T}$ is invertible?